

Bayesian data partitioning

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1 Introduction

Data partitioning is the problem of assigning each data vector $\mathbf{x}_i$, $i = 1, \dots, n$, a partition index $c_i \in \{1, \dots, C\}$. Classical approaches to data partitioning either define a cost function over the c_i and try to minimize it, or define the optimal partitioning operationally through an algorithm. This paper considers an alternative, model-based approach to data partitioning, in which the desired shape of the partitions is made explicit.

Denote each data partition by

$$D_j = \{\mathbf{x}_i | c_i = j\} \quad (1)$$

In this Bayesian framework, the data in D_j is assumed to be independently distributed according to a probability model $p(\mathbf{x}|\theta_j)$ with unknown parameters θ_j . Let $\theta = \{\theta_1, \dots, \theta_C\}$. The probability of a given partition $\mathbf{c}$ is proportional to

$$p(\mathbf{c}|D) \propto p(\mathbf{c}) \int_{\theta} \prod_i p(\mathbf{x}_i | \theta_{c_i}) p(\theta) d\theta \quad (2)$$

$$= p(\mathbf{c}) \int_{\theta} \prod_j p(D_j | \theta_j) p(\theta) d\theta \quad (3)$$

$$= p(\mathbf{c}) \prod_j \int_{\theta_j} p(D_j | \theta_j) p(\theta_j) d\theta_j \quad (4)$$

The last line follows when the partition parameters are independent under the prior: $p(\theta) = \prod_j p(\theta_j)$.

The prior probability on partitions $p(\mathbf{c})$ can be chosen in various ways. A uniform prior will prefer balanced clusters. A more flexible choice is a Polya distribution:

$$p(\mathbf{c}) = \int_{\mathbf{w}} \prod_j w_j^{n_j} p(\mathbf{w}) d\mathbf{w} \quad (5)$$

where $p(\mathbf{w})$ is Dirichlet, such as $\mathcal{D}(1, \dots, 1)$.

In this paper, the optimal partition is assumed to be the one with highest probability. Note that if all our priors are proper then the same approach can be used to select among varying numbers of partitions, i.e. varying C .

2 Gaussian partitions

As an example of the approach, suppose the partition model is Gaussian:

$$p(\mathbf{x}|\theta_j) = \mathcal{N}(\mathbf{x}; \mathbf{m}_j, \mathbf{V}_j) \quad (6)$$

Let n_j denote the size of D_j , $\bar{\mathbf{x}}_j$ denote the sample mean of D_j , and $\mathbf{S}_j$ denote the sample scatter of D_j .

Start by assuming the $\mathbf{V}_j$ are known. Then we have (assuming $n_j > 0$)

$$p(\mathbf{m}_j|\mathbf{V}_j) \sim \mathcal{N}(\mathbf{m}_0, \mathbf{V}_j/k_0) \quad (k_0 \rightarrow 0) \quad (7)$$

$$L_j = \int_{\theta_j} p(D_j|\theta_j)p(\theta_j)d\theta_j \quad (8)$$

$$= \int_{\mathbf{m}_j} p(D_j|\mathbf{m}_j, \mathbf{V}_j)p(\mathbf{m}_j|\mathbf{V}_j)d\mathbf{m}_j \quad (9)$$

$$= \frac{(\text{const.})}{n_j^{d/2} |2\pi \mathbf{V}_j|^{n_j/2}} \exp\left(-\frac{1}{2}\text{tr}(\mathbf{S}_j \mathbf{V}_j^{-1})\right) \quad (10)$$

$$-\sum_j 2\log L_j = d \sum_j \log n_j + n \sum_j \log |2\pi \mathbf{V}_j| + \sum_j \text{tr}(\mathbf{S}_j \mathbf{V}_j^{-1}) \quad (11)$$

When $\mathbf{V}_j = \mathbf{I}$, we want to minimize

$$J(\mathbf{c}) = \sum_j \sum_{i|c_i=j} \|\mathbf{x}_i - \bar{\mathbf{x}}_j\|^2 + d \sum_j \log n_j - 2 \log p(\mathbf{c}) \quad (12)$$

The first term is the same objective as k-means, and the second favors unbalanced partitions. Note that the third term doesn't cancel the second, since the prior for $\mathbf{c}$ cannot realistically depend on d . This objective desires spherical clusters of equal width, but not necessarily equal size, whereas k-means by itself prefers clusters to be spherical and balanced. If the data consists of two separated clusters with equal variance, but one has ten times as many points, then this objective will keep them separate while k-means will tend to grab data from the larger cluster in order to balance them out.

An EM algorithm to minimize J (when $p(\mathbf{c})$ is constant) is

E-step

$$\mathbf{m}_j = \bar{\mathbf{x}}_j \quad (13)$$

M-step

$$c_i = \underset{j}{\text{argmin}} \|\mathbf{x}_i - \mathbf{m}_j\|^2 + d/n_j \quad (14)$$

It is like k-means but penalizes small clusters according to d/n_j .

We can also use agglomeration. When we merge two clusters $j = 1$ and $j = 2$ the increase in cost is

$$\Delta_{12} = \mathbf{S}_{12} - \mathbf{S}_1 - \mathbf{S}_2 + d \log(n_1 + n_2) - d \log n_1 - d \log n_2 \quad (15)$$

$$= \frac{n_1 n_2}{n_1 + n_2} \|\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2\|^2 + d \log \frac{n_1 + n_2}{n_1 n_2} \quad (16)$$

Pick the two clusters which minimize Δ , merge them, and repeat.

Now let both $(\mathbf{m}_j, \mathbf{V}_j)$ be unknown:

$$p(\mathbf{V}_j) \sim \mathcal{W}^{-1}(\mathbf{S}_0, n_0) \quad (17)$$

$$L_j = (\text{const.}) \frac{\Gamma_d((n_j + n_0)/2)}{n_j^{d/2} \pi^{n_j d/2}} |\mathbf{S}_j + \mathbf{S}_0|^{-(n_j + n_0)/2} \quad (18)$$

Now we want to minimize

$$J(\mathbf{c}) = \sum_j (n_j + n_0) \log |\mathbf{S}_j + \mathbf{S}_0| + d \sum_j \log n_j - 2 \sum_j \log \Gamma_d((n_j + n_0)/2) - 2 \log p(\mathbf{c}) \quad (19)$$

The lead term is a novel variation on k-means-type objectives. It would be hard to interpret without the model-based derivation. With the derivation, we see that it prefers ellipsoidal clusters whose covariance matrix is similar to $\mathbf{S}_0/n_0$. An example prior is $\mathbf{S}_0 = d\mathbf{I}, n_0 = d$.

An EM algorithm to minimize J is

E-step

$$\mathbf{m}_j = \bar{\mathbf{x}}_j \quad (20)$$

$$\mathbf{V}_j = (\mathbf{S}_j + \mathbf{S}_0)/(n_j + n_0) \quad (21)$$

M-step

$$c_i = \underset{j}{\operatorname{argmin}} (\mathbf{x}_i - \mathbf{m}_j)^T \mathbf{V}_j^{-1} (\mathbf{x}_i - \mathbf{m}_j) + \log |(n_j + n_0) \mathbf{V}_j| + d/n_j - \Psi_d((n_j + n_0)/2) \quad (22)$$

3 Spherical data

As another example, suppose the data lives on the sphere, so a natural partition model is the Fisher/von Mises distribution:

$$p(\mathbf{x}|\theta_j = (\mathbf{m}_j, b_j)) = \frac{1}{Z_d(b_j)} \exp(b_j \mathbf{x}^\top \mathbf{m}_j) \quad (23)$$

$$Z_d(x) \propto \frac{I_{d/2-1}(x)}{x^{d/2-1}} \approx \frac{\exp(x)}{x^{(d-1)/2}} \quad (24)$$

Under this distribution, $E[\mathbf{x}] = \frac{Z'_d(b_j)}{Z_d(b_j)} \mathbf{m}_j \approx \mathbf{m}_j$. If the concentration parameter b_j is known, then with a uniform prior on $\mathbf{m}_j$:

$$L_j = \int_{\|\mathbf{m}_j\|=1} \frac{1}{Z_d(b_j)^{n_j}} \exp(b_j n_j \bar{\mathbf{x}}_j^\top \mathbf{m}_j) d\mathbf{m}_j \quad (25)$$

$$= \frac{Z_d(b_j n_j \|\bar{\mathbf{x}}_j\|)}{Z_d(b_j)^{n_j}} \quad (26)$$

If the concentration of all partitions is the same ($b_j = b$), then the cost function to minimize is

$$J(\mathbf{c}) = - \sum_j \log Z_d(b n_j \|\bar{\mathbf{x}}_j\|) \quad (27)$$

To make sense of this, note that

$$n_j \|\bar{\mathbf{x}}_j\| = \left\| \sum_i \mathbf{x}_i \right\| \quad (28)$$

$$\left\| \sum_i \mathbf{x}_i \right\|^2 = \sum_i \|\mathbf{x}_i\|^2 + \sum_{i \neq j} \mathbf{x}_i^\top \mathbf{x}_j \quad (29)$$

$$= n + \sum_{i \neq j} \mathbf{x}_i^\top \mathbf{x}_j \quad (30)$$

This is a measure of compactness on the sphere: when it is large, the vectors have large pairwise inner products, i.e. they have similar angle.

Despite the complicated-looking cost function, the EM algorithm is simple:

E-step

$$\mathbf{m}_j = \bar{\mathbf{x}}_j \quad (31)$$

M-step

$$c_i = \operatorname{argmax}_j \frac{\mathbf{x}_i^\top \mathbf{m}_j}{\|\mathbf{m}_j\|} \frac{Z'_d(b n_j \|\mathbf{m}_j\|)}{Z_d(b n_j \|\mathbf{m}_j\|)} \quad (32)$$

4 Other models

Bayesian objectives can also be easily derived for a multinomial model. See “Using lower bounds to approximate integrals”. One can even use nonparametric models for the partitions.